



Christopher Albert <chr.albert@gmail.com>

Re: Questions on an idea regarding misleading material conditional

5 Nachrichten

Graham Priest <priest.graham@gmail.com>

Di. 18. Sept. 2018, 14:26

An: Christopher Albert <chr.albert@gmail.com>

Hi, Chris, I'm afraid that what you say don't really work. See the attached.

G

On Mon, Sep 17, 2018 at 10:11 AM, Christopher Albert <chr.albert@gmail.com> wrote:

Dear Professor Priest,

I am a postdoc of theoretical physics with some background in mathematics. The (ab)use of language together with formal logics has always bugged me, so I drafted an explanation for myself and possibly others, how to deal with logical consequences alias implications.

The short version is that the material consequence $a \rightarrow b$ should be either spoken as "If a, then b, or a is false", or an ambiguous case of "irrelevance" should be introduced to represent "If a, then b" exactly via a different operation corresponding to a logical consequence. From a short literature search I found out that the latter change is related to paraconsistent logics (e.g. LP) and relevance logics, but I have not yet found an exact match of this interpretation.

Now I am really no expert in the field and know that you are kind of a "guru" in logics, so I wanted to ask you if you could take a quick look if what I wrote makes sense, recommend some literature on this specific topic, and answer a few questions. I hope this will take you only a very brief amount of time but it could help me save a lot. So for now the most- important questions:

1) Is the approach I am describing just the same as relevance logics or logics of paradox, or might there be more to it? Is there any equivalent in intuitionist logic or some other non-classical logic?

2) For a statement a I would like to distinguish

- * a
- * a is true
- * a is false

Are the latter two also considered as statements, or is there a better name for them?

3) Do you see any embarrassing misuse of words, letters, brackets etc. that hurt the eye of an expert immediately?

I would be very grateful if you could help me out on this, or indicate where to find further information.

All the best,
Christopher Albert

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Christopher Albert <chr.albert@gmail.com>
An: <priest.graham@gmail.com>

Di. 18. Sept. 2018, 14:53

Hi Graham,

thanks a lot for your comments. If I understand correctly, I have mixed up connectives and relations, which has led to the confusion. Adding the "I" does not solve the issue that the relation is decoupled from the truth value of a and b to avoid relating unrelated things. A good lesson that there is much more to logic than one would naively think.

Best,
Chris
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Graham Priest <priest.graham@gmail.com>
An: Christopher Albert <chr.albert@gmail.com>

Di. 18. Sept. 2018, 15:40

Yep. That seems right, Chris.

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Christopher Albert <chr.albert@gmail.com>
An: <priest.graham@gmail.com>

Mi. 19. Sept. 2018, 10:36

Hi Graham,

Thanks again and one more remark: I've now looked a bit how things are usually done in propositional logics with models, realisability, etc. and as I understand it, while explosion doesn't occur on the level of the material conditional $\rightarrow$ as a connective, it relates to the level of logical inference in the following way:

- 1) G is a consequence of F, if every model of F is also a model of G
- 2) In particular, if F is unsatisfiable, G is a consequence of F by this definition
- 3) Together, this makes the consequence equivalent to $F \rightarrow G$ being a tautology

So due to 2) we include the unintuitive cases and finally can cause explosion. The problem disappears if we don't have a contradiction in our theory. For ways to mitigate the problem, even if contradictions exist, I will study your book and a bit type theory in parallel to get the hang of it.

Best,
Chris
[Zitierter Text ausgeblendet]

Graham Priest <priest.graham@gmail.com>
An: Christopher Albert <chr.albert@gmail.com>

Mi. 19. Sept. 2018, 13:41

Hi, Chris. Yes, as I said,

$\models A \text{ hook } B \text{ iff } A \models B$

is true in classical logic, and some others. It fails in yet others.

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